

Effect of human H3N2 influenza virus reassortment on influenza incidence and severity during the 2017–18 influenza season in the USA: a retrospective observational genomic analysis

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Summary

Background During the 2017–18 influenza season in the USA, there was a high incidence of influenza illness and mortality. However, no apparent antigenic change was identified in the dominant H3N2 viruses, and the severity of the season could not be solely attributed to a vaccine mismatch. We aimed to investigate whether the altered virus properties resulting from gene reassortment were underlying causes of the increased case number and disease severity associated with the 2017–18 influenza season.

Methods Samples included were collected from patients with influenza who were prospectively recruited during the 2016–17 and 2017–18 influenza seasons at the Johns Hopkins Hospital Emergency Departments in Baltimore, MD, USA, as well as from archived samples from Johns Hopkins Health System sites. Among 647 recruited patients with influenza A virus infection, 411 patients with whole-genome sequences were available in the Johns Hopkins Center of Excellence for Influenza Research and Surveillance network during the 2016–17 and 2017–18 seasons. Phylogenetic trees were constructed based on viral whole-genome sequences. Representative viral isolates of the two seasons were characterised in immortalised cell lines and human nasal epithelial cell cultures, and patients' demographic data and clinical outcomes were analysed.

Findings Unique H3N2 reassortment events were observed, resulting in two predominant strains in the 2017–18 season: HA clade 3C.2a2 and clade 3C.3a, which had novel gene segment constellations containing gene segments from HA clade 3C.2a1 viruses. The reassortant re3C.2a2 viruses replicated with faster kinetics and to a higher peak titre compared with the parental 3C.2a2 and 3C.2a1 viruses (48 h vs 72 h). Furthermore, patients infected with reassortant 3C.2a2 viruses had higher Influenza Severity Scores than patients infected with the parental 3C.2a2 viruses (median 3.00 [IQR 1.00–4.00] vs 1.50 [1.00–2.00]; $p=0.018$).

Interpretation Our findings suggest that the increased severity of the 2017–18 influenza season was due in part to two intrasubtypes, cocirculating H3N2 reassortant viruses with fitness advantages over the parental viruses. This information could help inform future vaccine development and public health policies.

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Introduction

Influenza viruses threaten global public health through annual epidemics and less frequent pandemics. The error-prone replication of the viral genome introduces mutations that allow for the emergence of novel viruses to evade pre-existing, protective immunity in the human population (antigenic drift).¹ Further, these segmented genomes allow for reassortment events between influenza A viruses, resulting in progeny viruses with different genotypes. When co-circulation of multiple related lineages occurs, the more frequent intrasubtype reassortment between the viruses can increase the genomic diversity and adaptive potential.²

Intrasubtype reassortment events in H3N2 viruses have been observed in several historical epidemics,³ and they have resulted in the global spread of adamantane-resistant H3N2 viruses.⁴ Therefore, detailed whole-genome analyses paired with clinical data and virus characterisation can identify potential seasonal drifts or reassortment events that have public health consequences.

The 2016–17 and 2017–18 influenza seasons in the USA were dominated by H3N2 viruses. However, the 2017–18 season was characterised by a high disease burden, increased pneumonia and influenza mortality, and increased hospitalisations among the older population.⁵ The

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Research in context

Evidence before this study

We searched PubMed, with no language restrictions, for reports published from database inception to Dec 31, 2022, using the search terms “influenza virus” and “reassortment” and “disease severity”. We identified only one study that performed whole-genome analysis of reassortant influenza viruses, indicating that these viruses might possess the potential to cause human disease. However, this study did not include clinical data from patients in correlation with the viral genomic analysis. H3N2 viruses dominated the 2016–17 and 2017–18 influenza seasons in the USA, with the latter season having high rates of illness and mortality. Previous studies suggested that egg-adapted mutations in influenza vaccines during manufacturing could result in poor serum reactivity with recent H3N2 virus strains. Antigenic drift in H3N2 viruses was also thought to contribute to the severity of the 2017–18 season. However, there is a lack of evidence incorporating both virological and clinical information regarding this season.

Added value of this study

In this multicentre prospective and retrospective observational study, we found that the 2017–18 reassortants re3C.2a2 virus showed a replication advantage over parental viruses in primary differentiated human nasal epithelial cell cultures. This replication

advantage of reassortant viruses might contribute to increased severity of the infection, such as pneumonia and hospitalisations. In contrast, the 2017–18 reassortant re3C.3a virus introduced a glycosylated neuraminidase into the 3C.3a clade lineage, which enabled the virus to bypass pre-existing N2 NA-specific immunity and presumably infect more individuals. To our knowledge, this is currently the largest study that integrates phylogenetic, virological, and clinical data, offering evidence that influenza intrasubtype reassortant viruses might have the potential to cause human disease.

Implications of all the available evidence

We demonstrated that the ability to identify influenza-positive patients coupled with virus genome sequencing could provide detailed information and situational awareness of the influenza seasons. Our findings suggest that the increased severity of the 2017–18 influenza season might have been caused by the enhanced viral fitness of the cocirculating H3N2 reassortant viruses within the same subtype. Specifically, the reassortant re3C.2a2 viruses were associated with more severe influenza infections, while the reassortant re3C.3a viruses showed a higher ability to infect individuals aged 18 years and older in our study population.

H3N2 component, A/Hong Kong/4801/2014 (clade 3C.2a), of the influenza vaccine used in the northern hemisphere did not change over this period, nor was there an emergence of a clear antigenic drift variant H3N2. H3N2 clades in the 3C.2a lineage dominated in both seasons, but there was a low circulation of the antigenically distinct clade 3C.3a viruses.

The Johns Hopkins Center of Excellence for Influenza Research and Surveillance (JHCEIRS) network applied uniform criteria for influenza case detection that captures influenza-infected patients to perform in-depth genome sequencing and analysis of human influenza A virus. We hypothesise that altered virus properties resulting from gene reassortment were underlying causes of the increased case number and disease severity associated with the 2017–18 influenza season.

Methods

Study design

In this observational genomic study, we did a phylogenetic analysis of influenza A samples. These samples were collected from patients with influenza who were prospectively recruited during the 2016–17 and 2017–18 influenza seasons at the Johns Hopkins Hospital Emergency Departments in Baltimore, MD, USA, as well as from archived samples from the same seasons from Johns Hopkins Health System sites, constituting a combination of prospective and retrospective study elements (prospective and retrospective arms of the study). Our methods included whole-genome sequencing, infectious virus quantification, and neuraminidase inhibition assays. The analyses of these

data were done at Chang Gung University (Taoyuan, Taiwan) between Sept 1, 2022, and Feb 28, 2023.

The human subjects research protocol was reviewed and approved by the Johns Hopkins School of Medicine Institutional Review Board (IRB), which allowed for prospective screening, enrolment, and collection of samples from symptomatic patients with influenza at the Johns Hopkins Hospital (IRB00052743 and IRB00091667). We obtained written informed consent from participants in the prospective arm of the study. For the retrospective arm, the IRB waived the requirement for informed consent. We wrote this Article following the guidance of Strengthening The Reporting of Observational Studies in Epidemiology (STROBE) Statement (appendix pp 19–20).

Prospective enrolment, sample, and data collection

Patients presenting to the Johns Hopkins Hospital Emergency Departments (Baltimore, MD, USA) during the autumn and winter seasons (from Nov 1, 2016, to April 30, 2017, and from Nov 1, 2017, to April 30, 2018) were screened at triage using a clinical decision guideline that triggered a prompt to order an influenza test for patients presenting with a fever and any of three influenza-like illness symptoms (ie, cough, headache, and sore-throat). If symptoms were present, a rapid molecular diagnostic test for respiratory syncytium virus (RSV) and influenza A and B (Xpert Flu/RSV Assay Cepheid, Sunnyvale, CA, USA) was ordered, and a clinical coordinator was available to approach the patient to obtain informed consent and specimens as well as to conduct interviews to obtain clinical data for 16 h daily,

See Online for appendix.

from 0800 h to 0000 h, 7 days a week throughout the influenza seasons. Specimens (nasopharyngeal washes [NWS], nasopharyngeal swabs [NP]), and clinical data (demographics, ethnicity, medical and vaccination history, and symptoms) were collected at the time of enrolment (initial presentation).

Retrospective enrolment, sample, and data collection

We used residual nasopharyngeal wash or swab specimens (NWS or NP) collected for the purpose of respiratory virus testing and stored at the clinical laboratories in the Johns Hopkins Health System sites in the US National Capital Region and Baltimore during the same enrolment dates of the 2016–17 and 2017–18 seasons as those in the prospective arm. The clinical laboratories included in the retrospective arm were based at Johns Hopkins Hospital (Baltimore, MD, USA), Johns Hopkins Bayview Medical Center (Baltimore, MD, USA), Howard County General Hospital (Columbia, MD, USA), Sibley Memorial Hospital (Washington, DC, USA), and Suburban Hospital (Bethesda, MD, USA). Specimens were eligible for enrolment if the patients were aged 0–100 years of age, had positive influenza test results, and had more than 1 mL residual volume. Associated demographics and clinical data were collected via a review of the electronic health record. Specimens that were eligible for enrolment were chosen based on matching the age, gender, and hospital location of the patient between the two influenza seasons. Patients who were enrolled previously were excluded in both retrospective and prospective arms.

Covariates of interest and clinical outcomes

In our research, we considered several covariates to account for potential confounding factors and to assess their association with the clinical outcomes. The specific covariates we examined were: age, male gender, influenza vaccination, comorbidities, immunosuppressive medications, pregnancy, viral coinfection, days of illness, fever, cough, cough sputum, increased sputum, headache, sore throat, body aches, rhinorrhoea, shortness of breath, sinus pain, wheezing, fatigue, difficult getting out of bed, chest pain, chest hurt, shaking chills, appetite, nausea, diarrhoea, stomach pain, and conjunctivitis.

Our analysis focused on assessing the correlation between viral reassortments and clinical outcomes, including pneumonia development, physiological severity, and influenza severity. To measure the severity of influenza infection, we used two specific scoring systems: the National Early Warning Score (NEWS) and an Influenza Severity Score. NEWS is a widely used system that helps assess and monitor a patient's physiological parameters for early detection of clinical deterioration based on respiratory rate, oxygen saturation, any supplemental oxygen, temperature, systolic blood pressure, heart rate, and AVPU (alert, voice, pain, unresponsive) score. The Influenza Severity Score assists in gauging the severity of influenza in the studied population on the basis of age older than 45 years, male gender, number

of comorbidities, pneumonia, dyspnoea (shortness of breath), and confusion.⁶ Pneumonia also serves as another phenotype that patients infected with influenza might develop.

Virus whole-genome sequencing of clinical samples and virus isolates

Viral RNA of clinical samples and virus isolates was extracted following manufacturer instructions. Target amplicons were then whole-genome sequenced using Illumina kits and sequencers (appendix pp 1–2). The consensus sequence was generated based on the most likely subtype and used for further phylogenetic analyses.

Phylogenetic analysis

The initial maximum likelihood (ML) tree was constructed using FastTree software (version 2.1.11) with the Jukes-Cantor model and an estimate of 20 per-site rate categories. An intermediate ML tree was made from the initial results using RAxML and the rapid-hill climbing mode, a generalised time-reversible (GTR) model, and an estimate of 25 per-site rate categories, with the resulting tree evaluated under the GAMMA model of rate heterogeneity model. The final ML tree was made by optimising the model and branch lengths of the intermediate RAxML tree using RAxML with a GTR model, GAMMA model of rate heterogeneity, and ML estimate of the alpha-parameter (appendix p 2).

Eight viral gene segments were concatenated to generate one single sequence to make a whole-genome tree using A/Victoria/361/2011 as a root. Influenza virus clade was determined by the HA segment according to Global Influenza surveillance. Gene segments other than HA were assigned a clade on the basis of their tree branches to cluster with HA clades of 2016–17 parental viruses in the individual gene tree. Tips of the whole-genome tree were coloured by HA clade. The phylogenetic tree and heatmap of clades of individual gene segments were visualised using ggtree.

Infectious virus quantification and neuraminidase inhibition assay

We used plaque assays to determine the ability of the virus to infect and spread to neighbouring cells. The 50% tissue culture infectious dose (TCID₅₀) assays were used to quantify the replication of infectious viruses on MDCK-SIAT1 cells and human nasal epithelial cell cultures (described previously with some modifications; appendix p 3).

We used enzyme-linked lectin assays to quantify neuraminidase inhibition (NAI) in our study. We also measured NAI with an N2 monoclonal antibody 235–1C02 that recognises amino acids around active sites. NAI was defined as antibody concentration that resulted in at least 50% inhibition of neuraminidase activity compared with the virus without antibodies (appendix p 3).

Statistical analysis of demographic and clinical data

When appropriate, we used χ^2 or Fisher's exact tests for categorical variables and the rank-sum test for continuous

variables between three viral genotypes (3C.3/3C.3a vs 3C.2a1/3C.2a2/3C.2a3 comparison, parental 3C.3a vs reassortant 3C.3a comparison, and parental 3C.2a2 vs reassortant 3C.2a2 comparison). The two-tailed significance was set at *p* value lower than 0.05. To adjust for possible confounding effects between patient demographics, comorbidities, and clinical outcomes, a propensity score-based overlap weighting method was applied. We applied those potential confounders to generate the propensity (or probability) for being infected by different viral genotypes described above by using machine learning-based modelling methods. Specifically, in this study, we compared three viral genotypes (3C.3/3C.3a vs 3C.2a1/3C.2a2/3C.2a3, parental 3C.3a vs reassortant 3C.3a, and parental 3C.2a2 vs reassortant 3C.2a2). We used ML-based modelling to generate the propensity of being infected by the specific genotype, (eg, 3C.3/3C.3a vs 3C.2a1/3C.2a2/3C.2a3) for further confounder adjustment.⁷ By using the overlap weighting method, patients with higher propensity will have lower weights in the final multivariable logistic regression model to adjust for confounding factors.⁸ Using the weights, we created pseudo-populations in which the weighted virus genotypes have similar distribution across these confounders. The overlap weighting method could further eliminate inappropriate influence induced by extremely high or low weights. The overlap weighting method is an extension from the Inverse Probability of Treatment Weighting (IPTW). The overlap weight is the normalised version of $PS \times (1-PS)$, which attains its maximum at 0.5 where the two groups overlap most. Overlap weight addresses the non-overlapping support problem and overcomes the “too small denominator” problem. In our study, we selected three confounders identified from the univariable analysis to compare different viral genotypes. We included them in the propensity score model: age, multiple comorbidities, and influenza vaccination. Additionally, we compared several popular machine-learning algorithms, including generalised linear model, recursive partitioning algorithm, random forest, and gradient-boosting machine (GBM), to estimate the propensity scores. We used the area under the receiver operating characteristics curve (AUROC) to determine the best model that could generate the best propensity score to predict the virus genotype. We used the GBM model with the best AUROC (0.865) to generate the propensity score for the overlap weighting. We calculated coefficients or odds ratio (OR) and 95% CIs of unadjusted and overlap weighting adjusted results using logistic regression analysis. All statistical analyses were performed with Stata 16, GraphPad Prism 8 software, and R 3.6 (appendix p 4).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

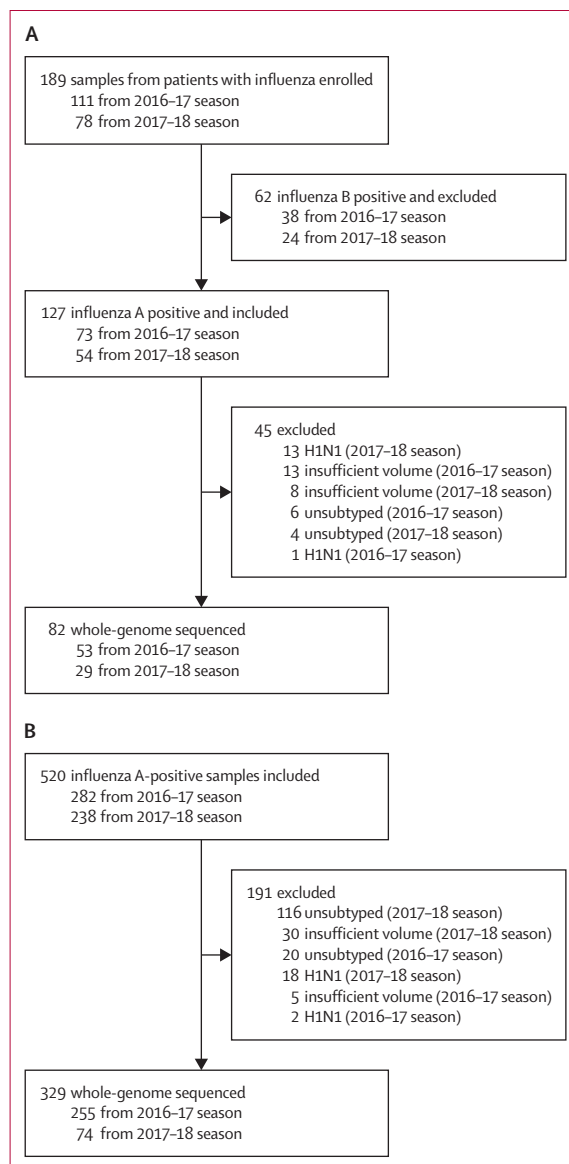


Figure 1: Flowchart of the 2016–17 and 2017–18 H3N2 surveillance study in Johns Hopkins Center of Excellence for Influenza Research and Surveillance network

Results

Consistent with influenza monitoring in the USA by the US Centers for Disease Control and Prevention (CDC), WHO, and other organisations, the 2017–18 influenza season in the Johns Hopkins Health System sites showed an earlier start and higher peak case numbers in the 2017–18 season ($n \sim 4600$) than in the 2016–17 season ($n \sim 3300$; appendix p 6, red line). A total of 647 samples from patients with influenza A virus infection were included in this study (figure 1).

An acute influenza case detection study prospectively enrolled 111 (from Nov 1, 2016, to April 30, 2017) and 78 (from Nov 1, 2017, to April 30, 2018) influenza-positive patients from the population seeking medical care at the

Johns Hopkins Hospital Emergency Departments (appendix p 8). Influenza A H3N2 viruses predominated in both seasons, with 90.4% (66 of 73) and 68.5% (37 of 54) identified in 2016–17 and 2017–18, respectively.

We determined HA clades for the majority of influenza A H3N2 samples from 2016–17 (90.9%, 60 of 66) and from 2017–18 (94.6%, 35 of 37) using HA gene sequencing data. Six samples from 2016–17 and two samples from 2017–18 were excluded because of poor sequencing results (appendix p 7). In 2016–17, the majority of H3N2 viruses (75.0%, 45 of 60) were clade 3C.2a1 (clade 3C.2a viruses with a 171Lys mutation), and the predominance was also seen in our retrospective surveillance and national data reported by the CDC in the USA throughout the season. H3N2 viruses from clade 3C.3a were present at higher numbers (20.0%, 12 of 60) than reported nationally by the CDC (7.3%). The rest of the H3N2 viruses were clade 3C.2a2 (5.0%, 3 of 60; defined as 3C.2a clade viruses with 131Lys/142Lys/261Gln). The dominant H3N2 HA clade in 2017–18 changed to 3C.2a2 (62.9%, 22 of 35), with only three (8.6%) of 35 samples identified as clade 3C.2a1 viruses (appendix p 7). H3N2 viruses from clade 3C.3a were also present at higher numbers (28.6%, 10 of 35) than those reported nationally by the CDC (6.6%) in 2017–18.

An additional 255 (2016–17) and 74 (2017–18) H3N2 samples from the retrospective surveillance system were sequenced for analysis (appendix p 11), and we found two more H3N2 HA clades (3C.2a3 [3.9%, ten of 255] in 2016–17 and 3C.3 [1.4%, one of 74] in 2017–18. Sequences were deposited to the Global Initiative on Sharing All Influenza Data (GISAID) database and the Johns Hopkins Data Repository (appendix pp 20–31).

The 2017–18 3C.2a2 genotype was a reassortant virus encoding the HA and PB1 segments from 2016–17 3C.2a2 and the remaining six segments from clade 3C.2a1, and this genotype also predominated in the 2017–18 influenza season in the USA (figure 2A, 2B, and 2C). The predominant 3C.3a genotype was a reassortant virus encoding the HA, M, and NS segments from 3C.3a and the remaining five segments from clade 3C.2a1. The 2017–18 season did have some 3C.3a clades that represented different reassortment events between 3C.3a, 3C.2a2, and 3C.2a1, but these represented a small percentage (4.9%, 5 of 103) of the total influenza samples sequenced. Taken together, our influenza genomic sequence surveillance identified reassortment in the two dominant H3N2 lineages circulating in the 2017–18 influenza season (figure 2D).

Positive virus isolations ($\text{TCID}_{50} > 3$) of viruses of each representative clade each season were passaged on MDCK-SIAT1 cells to generate virus stocks (appendix p 9). To determine whether 3C.2a2.2:6 reassortant virus had a replicative fitness advantage in vitro or ex vivo, two representative clinical isolates of each genotype 2016–17 3C.2a1, 2016–17 3C.2a2, and 2017–18 reassortant 3C.2a2 (re3C.2a2) were selected on the basis of their availability of isolation among the same genotype to compare their replication in vitro. The 2017–18 re3C.2a2

viruses produced more infectious viruses by less than a logarithmic unit in MDCK-SIAT1 cells than did the 2016–17 3C.2a2 or 3C.2a1 viruses after a low multiplicity of infection (MOI 0.001, figure 3). The 2017–18 re3C.2a2 viruses showed even faster kinetics and peak infectious virus titres after infection of hNEC cultures suggesting the reassortant viruses have a replication advantage over parental viruses in this physiologically relevant culture system (48 h vs 72 h, figure 3D). Overall, the 2017–18 re3C.2a2 increased replication kinetics in vitro and ex vivo over the parental 3C.2a2 and 3C.2a1 viruses.

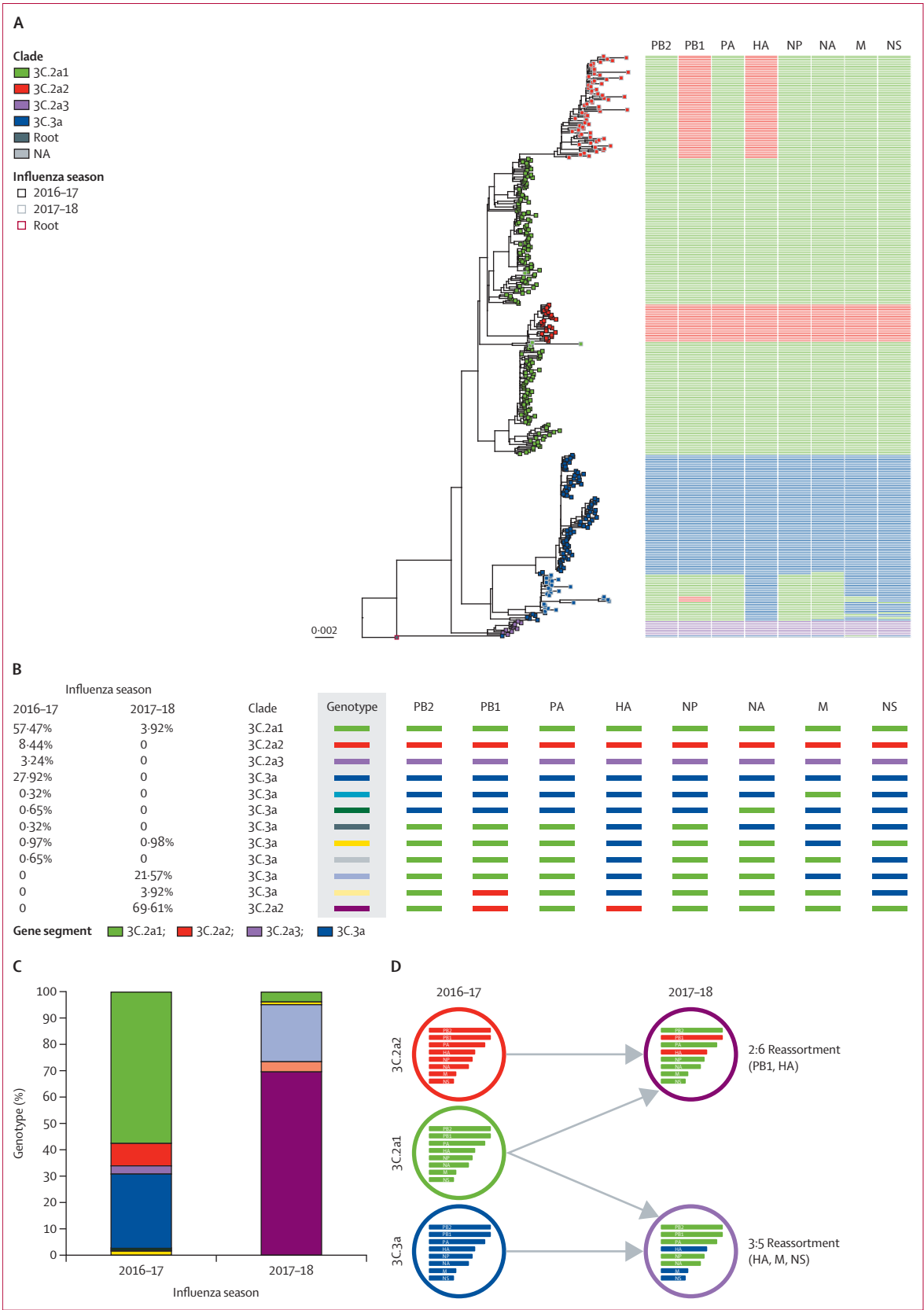
There was no replication advantage of re3C.3a virus compared with parental 3C.3a (figure 3E, 3F, 3G and 3H). A monoclonal antibody targeting the neuraminidase active site was able to effectively and, in a dose-dependent manner, inhibit the neuraminidase activity of 2016–17 3C.3a viruses, but it did not inhibit neuraminidase activity of 2017–18 re3C.3a viruses (figure 3I).

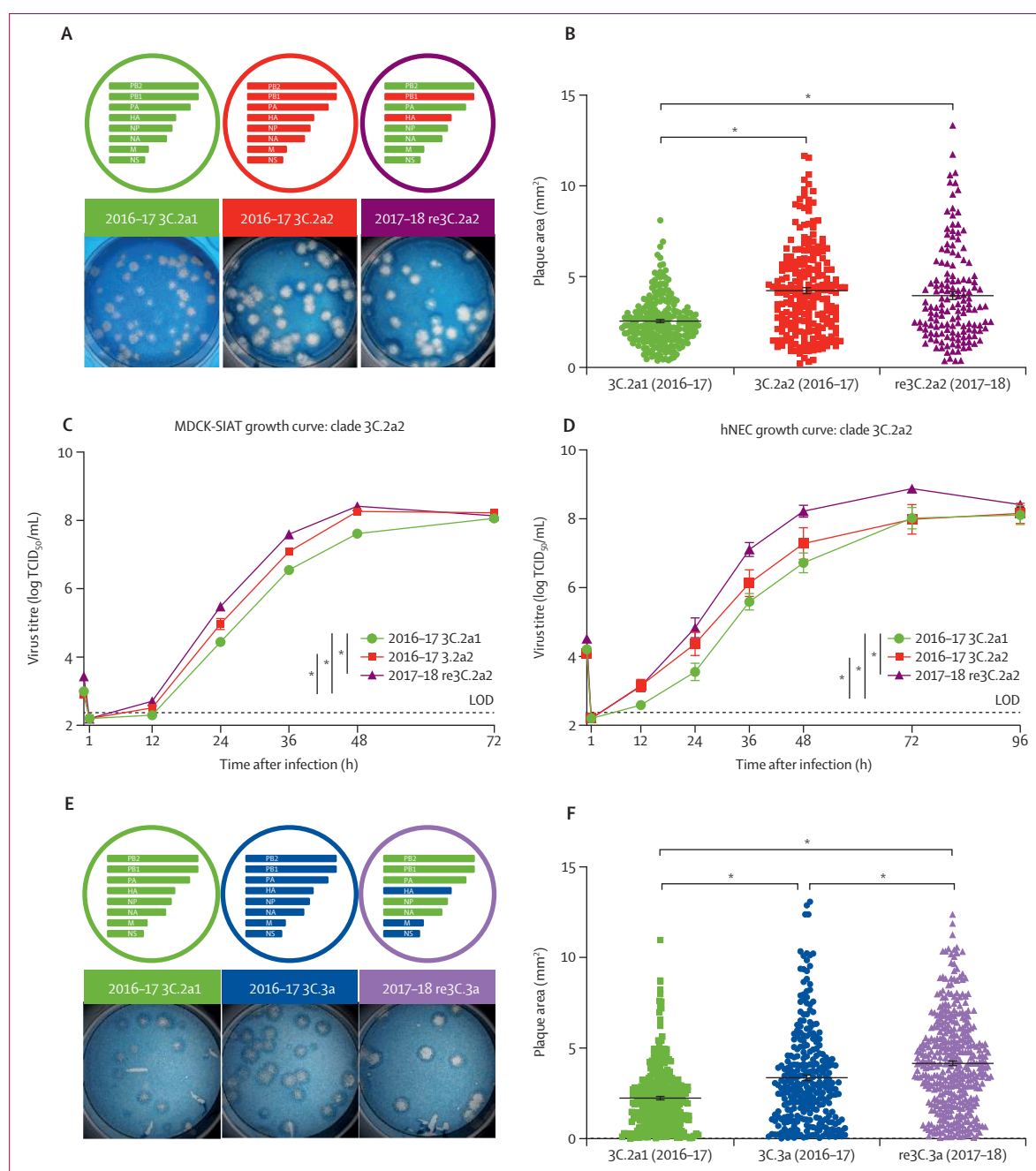
We then aimed to understand the potential correlation between different viral genotypes and patient demographics, clinical symptoms, and clinical outcomes. To do so, we analysed 411 whole-genome sequences from the 647 samples from influenza A-positive patients recruited. We checked the distributions of patient characteristics, symptoms, and severity but found no significant difference between patients with and without whole-genome sequence analysis (appendix p 12).

Patients infected with clade 3C.2a viruses, including 3C.2a1, 3C.2a2, or 3C.2a3 clades, were older, more likely to have comorbidities, cough with increased sputum and headache, less likely to have fever and running nose, and more likely to be admitted and diagnosed with pneumonia than patients infected with clades 3C.3 and 3C.3a (table 1 and appendix p 15). There were no differences in age, gender, comorbidity, or disease outcome between patients infected with 3C.2a1 clade viruses and those infected with 3C.2a2 clade viruses; the only difference was that 3C.2a1 clade was predominant in 2016–17, but 3C.2a2 clade was predominant in 2017–18 (appendix p 16). The distributions of possible confounders, such as age, comorbidities, and influenza vaccination, became similar between different virus genotypes after the machine-learning-based propensity-score overlapping weighting adjustment (appendix p 9). After we applied the machine-learning-based propensity-score overlapping weighting adjustment for potential confounding from age and comorbidities and influenza vaccination, the association between 3C.2a and pneumonia disappeared (OR 4.80, 95% CI 1.88–16.26 vs aOR 2.72, 95% CI 0.87–8.52; table 2 and appendix p 9).

Furthermore, patients infected with re3C.2a2 viruses had higher severity scores than patients infected with the parental 3C.2a2 viruses (median 3.00 [IQR 1.00–4.00] vs 1.50 [1.00–2.00], $p=0.018$; table 3 and appendix p 14). However, there was no difference in age, gender, number of comorbidities, or pneumonia between the two genotypes. The higher severity score was highly likely due to a higher proportion of patients infected with re3C.2a2 viruses having

Figure 2: Analysis of circulating H3N2 viruses in the 2016–17 and 2017–18 influenza seasons in the US National Capital Region and Baltimore with reassortant viruses
(A) Whole-genome phylogenetic tree. Maximum likelihood trees of concatenated gene segments of H3N2 viruses in influenza prospective and retrospective surveillance samples at Johns Hopkins Health System during 2016–18. Tips are coloured by HA clade and influenza season. The heatmap shows the assigned clade of each gene corresponding to its genome on the tree. One patient infected with 3C.3 clade viruses was not included.
(B) Genotypes and (C) percentage of genotypes of H3N2 viruses circulated in US National Capital Region during the 2016–17 and 2017–18 seasons.
(D) Summary of the 2:6 and 3:5 intra-subtype reassortment events between the H3N2 3C.2a2 or 3C.3a and 3C.2a1 subclades, respectively.





(Figure 3 continues on next page)

shortness of breath (34 of 71 [47.9%]) than patients infected with the parental 3C.2a2 viruses (3 of 26 [11.5%], $p < 0.0001$; appendix p 13).

Additionally, patients infected with re3C.3a tended to be older, with more comorbidities and symptoms, were more likely to be vaccinated, and had higher severity scores than patients with parental 3C.3a (appendix p 16). The higher severity scores in re3C.3a patients were only observed in patients at age 18 years and older, not in paediatric patients

(appendix p 17). However, the association between re3C.3a and severity score became non-significant after adjusting the confounders for age, comorbidities, and vaccination status (table 2), suggesting that parental 3C.3a or re3C.3a viruses seem to not result in differential clinical outcomes.

Discussion

We report on targeted influenza surveillance and case detection in a single geographical region: the US National

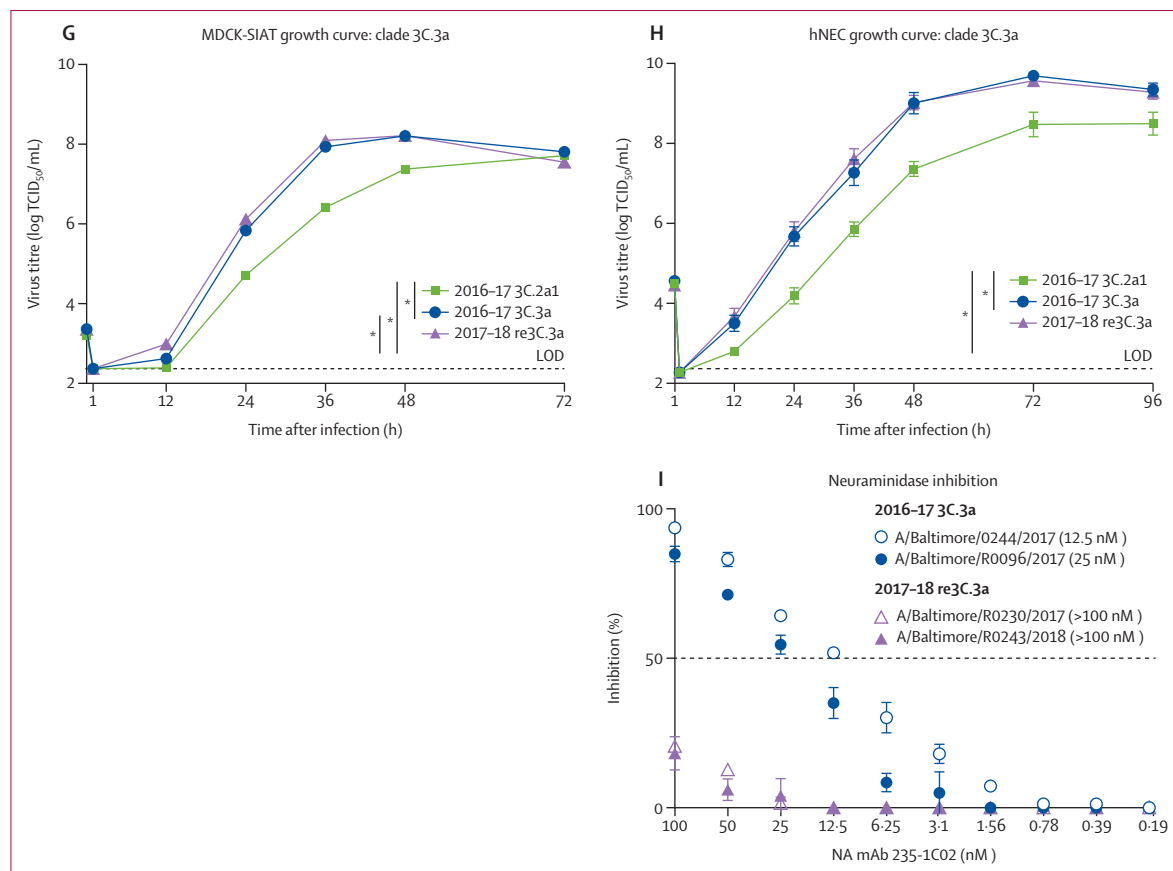


Figure 3: Viral fitness of reassortant viruses compared with non-reassortant viruses

Representative images of plaque assays on MDCK cells (A, E) and the quantification of plaque area (B, F). Multistep growth curves on MDCK-SIAT1 cells (MOI 0.001 TCID₅₀ units per cell; C, G) and hNECs (MOI 0.01 TCID₅₀ units per cell; D, H). Data were from two or three independent experiments and mean (SEM) are shown. Dotted line indicates LOD. Statistical analyses were done by one-way ANOVA for plaque assays and by two-way ANOVA for growth curves. (I) NAI assays were performed with N2 monoclonal antibody (235-1C02) that recognises amino acids around the active site. Each virus isolate was graphed from duplicates. Mean (SEM) were shown. hNECs=human nasal epithelial cells. LOD=limit of detection. MOI=multiplicity of infection. NAI=neuraminidase inhibition. SEM=standard error of mean. TCID₅₀= median tissue culture infectious dose. **p*<0.05.

Capital Region and Baltimore. Identifying influenza-positive patients and determining viral genome sequences provides insights into how reassortment events influence the influenza season. With detailed clinical information, we found the reassortant re3C.2a2 viruses could be attributed to more severe influenza infection while reassortant re3C.3a viruses were able to infect more individuals aged 18 years and older in this study population compared with their parental 3C.3a viruses. To our knowledge, two intra-subtype reassortant viral strains dominantly co-circulating in the same geographical area in the same season have not been documented previously.

The H3N2 influenza vaccine component in the 2016–17 and 2017–18 northern hemisphere formulation was the 3C.2a clade virus A/Hong Kong/4801/2014.⁹ H3N2 vaccine effectiveness in each of these seasons was approximately 25%.^{10,11} A diverse group of H3N2 clades co-circulated not only in the US National Capital Region, but in the USA and worldwide during these seasons.^{9,12} Ferret serum raised

from the egg-culture-adapted A/Hong Kong/4801/2014 virus neutralised poorly with 50% of circulating 3C.2a1 or 3C.2a2 viruses from the 2017–18 season, but reactivity of serum raised to a cell culture propagated A/Hong Kong/4801/2014 variant was significantly higher.⁵ It was suggested that egg-adapted mutations obtained through manufacturing were the reason behind the poor serum reactivity with recent H3N2 virus strains.^{13–15} 3C.3a viruses are antigenically distinct from 3C.2a viruses.^{5,16} The 2016–17 and 2017–18 H3N2 influenza vaccine formulations produced little protective immunity to 3C.3a viruses. Serological testing has indicated that the 3C.2a2 viruses are antigenically indistinguishable from cell-culture-propagated 3C.2a1 viruses.¹⁷ These data indicate that the increased number of influenza cases in 2017–18 cannot be attributed to a classic antigenic drift of circulating viruses or a poorly immunogenic vaccine alone. The enhanced in-vitro replication of re3C.2a2 viruses could indicate that an overall increase in virus replication fitness led to increased

	Overall (n=411)	3C.2a1 (n=181)	3C.2a2 (n=97)	3C.2a3 (n=10)	3C.3 + 3C.3a (n=123)
Flu season 2016-17					
Total	308 (75%)	177 (98%)	26 (27%)	10 (100%)	95 (77%)
Prospective	53 (13%)	40 (22%)	3 (3%)	0	10 (8%)
Retrospective	255 (62%)	137 (76%)	23 (24%)	10 (100%)	85 (69%)
Flu season 2017-18					
Total	103 (25%)	4 (2%)	71 (73%)	0	28 (23%)
Prospective	29 (7%)	1 (1%)	19 (20%)	0	9 (7%)
Retrospective	74 (18%)	3 (2%)	52 (54%)	0	19 (15%)
Demographics					
Age					
Median	43.0 (10.0-76.0)	68.00 (23.0-82.0)	62.00 (21.0-81.0)	23.00 (13.8-77.5)	9.00 (5.0-28.5)
<18 years	145 (35%)	32 (18%)	22 (23%)	4 (40%)	87 (71%)
18-49 years	73 (18%)	36 (20%)	20 (21%)	2 (20%)	15 (12%)
50-64 years	33 (8%)	21 (12%)	8 (8%)	0	4 (3%)
≥65 years	160 (39%)	92 (51%)	47 (49%)	4 (40%)	17 (14%)
Gender					
Male	190 (46%)	80 (44%)	48 (50%)	7 (70%)	55 (45%)
Female	211 (53%)	101 (56%)	49 (50%)	3 (30%)	68 (55%)
Received influenza vaccine	150 (37%)	67 (37%)	35 (36%)	3 (30%)	45 (37%)
Ethnicity					
American Indian or Alaska Native	1 (<1%)	0	0	0	1 (1%)
Asian	21 (5%)	11 (6%)	4 (4%)	0	6 (5%)
Native Hawaiian or other Pacific Islander	1 (<1%)	0	0	0	1 (1%)
African American	194 (47%)	77 (43%)	39 (40%)	3 (30%)	75 (61%)
White	162 (39%)	85 (47%)	44 (45%)	7 (70%)	26 (21%)
Other	31 (8%)	8 (4%)	9 (9%)	0	14 (11%)
Comorbidities					
No comorbidities	146 (36%)	56 (31%)	29 (30%)	3 (30%)	58 (47%)
One comorbidity	61 (15%)	29 (16%)	20 (21%)	3 (30%)	9 (7%)
Multiple comorbidities	204 (50%)	96 (53%)	48 (50%)	4 (40%)	56 (46%)
Immunosuppressive medication	3 (1%)	2 (1%)	1 (1%)	0	0
Disease severity					
NEWS score	2.0 (0.0-4.0)	2.0 (1.0-4.0)	2.00 (0.0-4.0)	1.5 (0.3-3.0)	2.0 (0.0-6.0)
Oxygen supplementation	24 (6%)	14 (8%)	5 (5%)	1 (10%)	4 (3%)
BiPAP or CPAP	0	0	0	0	0
Intubated	0	0	0	0	0
Admitted to hospital	103 (25%)	52 (29%)	31 (32%)	2 (20%)	18 (15%)
Length of hospital stay, days	2.0 (1.0-5.8)	2.0 (1.0-5.3)	4.0 (2.0-9.0)	N/A	2.5 (1.8-3.8)
Pneumonia	44 (11%)	21 (12%)	18 (19%)	1 (10%)	4 (3%)
Severity score	2.0 (1.0-3.0)	2.0 (1.0-3.0)	2.0 (1.0-4.0)	1.0 (1.0-2.8)	1.0 (0.0-2.0)
Intensive care unit	5 (1%)	4 (2%)	1 (1%)	0	0
Death	1 (<1%)	1 (1%)	0	0	0
Viral co-infection	19 (5%)	11 (6%)	4 (4%)	0	4 (3%)
Symptom					
Symptom duration, days*	2.0 (1.0-3.0)	2.0 (1.0-3.0)	1.5 (1.0-2.0)	1.0 (1.0-1.0)	2.0 (1.0-3.0)
Fever	318 (77%)	139 (77%)	71 (73%)	4 (40%)	104 (85%)
Cough	347 (84%)	156 (86%)	85 (88%)	9 (90%)	97 (79%)
Cough sputum	127 (31%)	62 (34%)	39 (40%)	1 (10%)	25 (20%)
Increased sputum	56 (14%)	30 (17%)	16 (17%)	1 (10%)	9 (7%)
Headache	134 (33%)	61 (34%)	37 (38%)	5 (50%)	31 (25%)
Sore throat	121 (29%)	51 (28%)	24 (25%)	4 (40%)	42 (34%)
Body aches	132 (32%)	60 (33%)	36 (37%)	4 (40%)	32 (26%)
Runny nose	214 (52%)	84 (46%)	46 (47%)	3 (30%)	81 (66%)
Shortness of breath	137 (33%)	63 (35%)	37 (38%)	3 (30%)	34 (28%)
Sinus pain	39 (10%)	20 (11%)	10 (10%)	0	9 (7%)

(Table 1 continues on next page)

	Overall (n=411)	3C.2a1 (n=181)	3C.2a2 (n=97)	3C.2a3 (n=10)	3C.3 + 3C.3a (n=123)
(Continued from previous page)					
Wheezing	98 (24%)	39 (22%)	27 (28%)	2 (20%)	30 (24%)
Fatigue	155 (38%)	72 (40%)	44 (45%)	1 (10%)	38 (31%)
Difficulty getting out of bed	76 (19%)	36 (20%)	21 (22%)	0	19 (15%)
Chest pain	95 (23%)	41 (23%)	23 (24%)	3 (30%)	28 (23%)
Chest hurt	48 (12%)	20 (11%)	15 (16%)	0	13 (11%)
Shaking chills	121 (29%)	59 (33%)	32 (33%)	1 (10%)	29 (24%)
Loss of appetite	126 (31%)	49 (27%)	35 (36%)	1 (10%)	41 (33%)
Nausea	131 (32%)	60 (33%)	31 (32%)	2 (20%)	38 (31%)
Diarrhoea	52 (13%)	22 (12%)	13 (13%)	1 (10%)	16 (13%)
Stomach pain	59 (14%)	23 (13%)	18 (19%)	1 (10%)	17 (14%)

Data are n (%) or median (IQR). NEWS=National Early Warning Score. N/A=not-available. BiPAP=bilevel positive airway pressure. CPAP=continuous positive airway pressure. *2.4% missingness, value expressed as mean and standard deviation unless noted.

Table 1: Demographic and clinical comparisons of patients infected with H3N2 clades in the US National Capital Region during 2016–17 and 2017–18 seasons

Outcome (Pneumonia)		Unadjusted		Overlap weight*	
		OR (95% CI)	p value	Adjusted OR (95% CI)	p value
3C.3 and 3C.3a	4 (3%)	1 (ref)	..	..	..
3C.2a1, 3C.2a2, and 3C.2a3	40 (14%)	4.80 (1.88 to 16.26)	0.003	2.72 (0.87 to 8.52)	0.086

Data are n (%) or OR (95% CI). OR=odds ratio. *Overlap weight results were done by propensity score weighting with the gradient-boosting machine (area under the receiver operating characteristic curve: 0.865) for age and comorbidities.

Table 2: Regression analysis in patients with H3N2 influenza in the US National Capital Region during 2016–17 and 2017–18 seasons by clade

Outcome (Influenza Severity Score)		Unadjusted		Overlap weight*	
		Coefficient (95% CI)	p value	Coefficient (95% CI)	p value
Parental 3C.3a	1.00 (0.00 to 2.00)	1 (ref)	..	..	..
Reassortant 3C.3a	2.00 (0.25 to 4.00)	0.79 (0.16 to 1.42)	0.016	0.25 (−0.29 to 0.80)	0.363
Parental 3C.2a2	1.50 (1.00 to 2.00)	1 (ref)	..	..	..
Reassortant 3C.2a2	3.00 (1.00 to 4.00)	0.82 (0.06 to 1.58)	0.038	..	..

Data are median (IQR) or coefficient (95% CI). *Overlap weight results were done by propensity score weighting with the gradient-boosting machine (area under the receiver operating characteristic curve: 0.865) for age, comorbidities, and vaccination status.

Table 3: Regression analysis in H3N2 patients in the US National Capital Region during 2016–17 and 2017–18 seasons by genotype

numbers of infections in 2017–18 irrespective of vaccination status, thus explaining the higher numbers of influenza cases with no significant decrease in vaccine efficacy.

Reassortments between influenza A virus subtypes circulating in humans and animals have been documented to occur with some frequency in animal populations^{18–20} and have resulted in viruses that have led to influenza pandemics such as those in 1957, 1968, and 2009.^{21,22} Intrasubtype reassortment between human influenza viruses has been documented, but its relationship to increased influenza disease is unclear. Intrasubtype reassortments of human H1N1 viruses were associated with unusually severe seasonal epidemics in 1947 and 1951.²³ Intrasubtype reassortment in 2002–03 might have contributed to the emergence of the H3N2 A/Fujian/411/2002-like antigenic drift virus responsible for the poor vaccine-match 2003–04 influenza season.²⁴ Another study of the H3N2 2014–15 season in the UK suggested that viral reassortment

correlated to severe disease and occurred later in the season.²⁵ Increasing disease severity could be due to mutations, but subsequent reassortment might have contributed to increased fitness of the virus in humans. Two studies determined an extensive presence of intrasubtype reassortment human H3N2 viruses circulating from 1968 to 2011 and from 2008 to 2014.^{3,26} However, reassortment of the neuraminidase segment between 3C.3a and 3C.2a clades was identified in the UK in the 2014–15 season,²⁵ but this reassortant virus did not become established in the human population. We detected the 3C.3a intrasubtype reassortant viruses in the 2016–17 season in the US National Capital region, but these did not outcompete the parental 3C.3a viruses in that season. Reassortments of 3C.2a2 and 3C.3a with 3C.2a1 internal segments circulated in the US National Capital Region in 2017–18 and probably in the USA. The identification of two distinct viruses in the same influenza subtype that has undergone reassortment in

the human population at the same time has yet to be documented previously.

Our data suggest that reassortment brought different properties to the 3C.2a2 and 3C.3a clade viruses even though the reassortment events occurred at the same time. The 3C.2a2 reassortment clearly improved virus replication in transformed cell lines and human nasal epithelial cell cultures. This increased replication could drive both increased disease severity and increased transmission in infected individuals, as both observations are affected by the overall virus load. In contrast, reassortment did not change the overall replication of the re3C.3a viruses compared with the parental 3C.3a viruses. In 2014–15, the clade 3C.2a H3N2 human influenza viruses acquired a mutation in the neuraminidase protein that resulted in glycosylation at amino acid 245.²⁷ This glycosylation alters the overall enzymatic activity of neuraminidase and masks antigenic sites for antibodies that inhibit neuraminidase enzymatic activity and neutralise virus infectivity.²⁷ The 2017–18 3C.3a reassortment event replaced a 245 neuraminidase glycosylation-negative protein with the 245 neuraminidase glycosylation-positive protein that was only seen in the 3C.2a lineage viruses until that time. Our data showed that the 2017–18 3C.3a reassortment event introduced a glycosylated neuraminidase into the 3C.3a clade lineage that allowed the virus to circumvent pre-existing N2 NA-specific immunity in a manner similar to what was demonstrated for post-2016 clade 3C.2a H3N2 viruses. The acquisition of a 245 neuraminidase glycosylation-positive protein by re3C.3a viruses allowed these viruses to evade some pre-existing anti-neuraminidase antibody immunity in the population,²⁷ thus increasing the number of susceptible individuals in the population from children to the middle-aged population and allowing this clade to continue to circulate in the USA in 2018–19 and in Europe in 2019–20.²⁸ Correlating clinical disease symptoms with reassortant virus infection in our surveillance provides supportive evidence for understanding the roles of the 2017–18 reassortant viruses. These viruses, characterised by enhanced replication and neuraminidase antibody immune evasion, might substantially influence clinical outcomes.

The dominance of reassortant re3C.2a2 and re3C.3a viruses in the 2017–18 influenza season is likely to indicate an increased fitness of these viruses compared with their parental HA clade viruses. Reassortant re3C.3a viruses became the dominant H3N2 viruses that circulated in 2018–19 in the US with some 3C.2a1b viruses, but the re3C.2a2 viruses did not continue to circulate.²⁸ The 2018–19 H3N2 vaccine A/Singapore/INFIMH-16-0019/2016 was a clade 3C.2a1 virus, which was antigenically distinct from the dominant clade 3C.3a in the USA that season.²⁸ It is not surprising that the vaccine effectiveness of all ages to H3N2 in the 2018–19 in the USA was only 9%,²⁹ and the 2018–19 H3N2 vaccine provided no protection among adults for hospitalisation.³⁰

Our study has some limitations. First, owing to the nature of the part of the prospective surveillance network, we might

only be capturing some influenza viruses circulating in the community. In addition, the virus circulation in one single metropolitan area might also only represent part of the population. However, the main research question we hope to answer in this study is the association between intra-subtype reassortment events and clinical outcomes. Accordingly, the possible bias of the surveillance effort would be minimal. Second, not all patients in this study could contribute to the whole-genome sequence analysis. However, we checked the distribution of virus clades, patient characteristics, and severity but found no statistical difference between patients with and without whole-genome sequence analysis. Third, the smaller sample sizes in the subgroup analyses prevent us from making a firm conclusion for the causal inference between the intra-subtype reassortment and clinical outcomes. This limited sample size also prevents us from confidently making the predominance statement of the circulating strains in these seasons. Lastly, we cannot assess the geospatial information from the patients included in this study.

Intense influenza surveillance that relies on whole-genome sequencing combined with phenotypic assessment of new virus lineages allowed us to identify new genetic changes that allowed continued virus circulation. Improved pathogen genomic sequencing needs to be a regular part of surveillance efforts to provide a more detailed understanding of the sometimes-subtle changes that occur in pathogens that can provide a fitness advantage and result in the increased spread or disease severity.

Contributors

HL, RER, and AP conceived of the experimental questions. KS-S, K-FC, and LMS oversaw enrolment and collection of samples from patients. TM, JE, PT, AD, and HL sequenced samples and cleaned the raw data. HL, JW, TM, and AWE performed phylogenetic tree analyses. HL, JW, DJ, HP, ZM, and SV isolated viruses and performed virus characterisation experiments. C-YC, C-CW, and K-FC ran demographic and clinical analyses. KS-S and S-HC accessed and verified data. HL, KS-S, JW, Y-NG, and K-FC analysed data. HL, KS-S, S-HC, K-FC, and AP wrote the manuscript. All authors edited and reviewed the manuscript before submission. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

SV owns stock in VIR Biotechnology as an employee of the company, unrelated to the submitted work. The other authors declare no competing interests.

Data sharing

Sequences were deposited to the Global Initiative on Sharing All Influenza Data (GISAID) database and the Johns Hopkins Data Repository (appendix pp 23–33). Clinical data, immunoassay raw data, and code are accessible upon request by emailing the corresponding authors.

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References

- Petrova VN, Russell CA. The evolution of seasonal influenza viruses. *Nat Rev Microbiol* 2018; **16**: 60.
- Lowen AC. Constraints, drivers, and implications of influenza A virus reassortment. *Annu Rev Virol* 2017; **4**: 105–21.
- Westgeest KB, Russell CA, Lin X, et al. Genomewide analysis of reassortment and evolution of human influenza A(H3N2) viruses circulating between 1968 and 2011. *J Virol* 2014; **88**: 2844–57.
- Simonsen L, Viboud C, Grenfell BT, et al. The genesis and spread of reassortment human influenza A/H3N2 viruses conferring adamantane resistance. *Mol Biol Evol* 2007; **24**: 1811–20.
- Garten R, Blanton L, Elal AIA, et al. Update: influenza activity in the United States during the 2017–18 season and composition of the 2018–19 influenza vaccine. *MMWR Morb Mortal Wkly Rep* 2018; **67**: 634–42.
- Capelastegui A, Quintana JM, Bilbao A, et al. Score to identify the severity of adult patients with influenza A (H1N1) 2009 virus infection at hospital admission. *Eur J Clin Microbiol Infect Dis* 2012; **31**: 2693–701.
- Lee BK, Lessler J, Stuart EA. Improving propensity score weighting using machine learning. *Stat Med* 2010; **29**: 337–46.
- Li F, Thomas LE, Li F. Addressing extreme propensity scores via the overlap weights. *Am J Epidemiol* 2019; **188**: 250–57.
- WHO. Recommended composition of influenza virus vaccines for use in the 2017–2018 northern hemisphere influenza season. *Wkly Epidemiol Rec* 2017; **92**: 117–28.
- Flannery B, Chung JR, Belongia EA, et al. Interim Estimates of 2017–18 Seasonal Influenza Vaccine Effectiveness – United States, February 2018. *MMWR Morb Mortal Wkly Rep* 2018; **67**: 180–85.
- Skowronski DM, Chambers C, De Serres G, et al. Early season co-circulation of influenza A(H3N2) and B(Yamagata): interim estimates of 2017/18 vaccine effectiveness, Canada, January 2018. *Euro Surveill* 2018; **23**: 18–00035.
- World Health Organization. Recommended composition of influenza virus vaccines for use in the 2018–2019 northern hemisphere influenza season. *Wkly Epidemiol Rec* 2018; **93**: 133–41.
- Wu NC, Zost SJ, Thompson AJ, et al. A structural explanation for the low effectiveness of the seasonal influenza H3N2 vaccine. *PLoS Pathog* 2017; **13**: e1006682.
- Ursin RL, Liu H, Powell HR, et al. Differential antibody recognition of H3N2 vaccine and seasonal influenza virus strains based on age, vaccine status, and sex in the 2017–2018 season. *J Infect Dis* 2020; **222**: 1371–82.
- Zost SJ, Parkhouse K, Gumina ME, et al. Contemporary H3N2 influenza viruses have a glycosylation site that alters binding of antibodies elicited by egg-adapted vaccine strains. *Proc Natl Acad Sci U S A* 2017; **114**: 12578–83.
- Blanton L, Alabi N, Mustaqim D, et al. Update: influenza activity in the United States during the 2016–17 season and composition of the 2017–18 influenza vaccine. *MMWR Morb Mortal Wkly Rep* 2017; **66**: 668–76.
- ECDC. Influenza virus characterisation, summary Europe, March 2018. ECDC Surveillance Report. Stockholm: European Centre for Disease Prevention and Control, 2018.
- Naguib MM, Ulrich R, Kasbohm E, et al. Natural reassortants of potentially zoonotic avian influenza viruses H5N1 and H9N2 from Egypt display distinct pathogenic phenotypes in experimentally infected chickens and ferrets. *J Virol* 2017; **91**: e01300–17.
- Wang H, Zhang Z, Chen Z, et al. High genetic diversity and frequent genetic reassortment of avian influenza A(H9N2) viruses along the East Asian-Australian migratory flyway. *Infect Genet Evol* 2016; **39**: 325–29.
- Ma L, Jin T, Wang H, et al. Two reassortant types of highly pathogenic H5N8 avian influenza virus from wild birds in Central China in 2016. *Emerg Microbes Infect* 2018; **7**: 14.
- Horimoto T, Kawaoka Y. Influenza: lessons from past pandemics, warnings from current incidents. *Nat Rev Microbiol* 2005; **3**: 591–600.
- Garten RJ, Davis CT, Russell CA, et al. Antigenic and genetic characteristics of swine-origin 2009 A(H1N1) influenza viruses circulating in humans. *Science* 2009; **325**: 197–201.
- Nelson MI, Viboud C, Simonsen L, et al. Multiple reassortment events in the evolutionary history of H1N1 influenza A virus since 1918. *PLoS Pathog* 2008; **4**: e1000012.
- Holmes EC, Ghedin E, Miller N, et al. Whole-genome analysis of human influenza A virus reveals multiple persistent lineages and reassortment among recent H3N2 viruses. *PLoS Biol* 2005; **3**: e300.
- Goldstein EJ, Harvey WT, Wilkie GS, et al. Integrating patient and whole-genome sequencing data to provide insights into the epidemiology of seasonal influenza A(H3N2) viruses. *Microb Genom* 2018; **4**: e000137.
- Maljkovic Berry I, Melendrez MC, Li T, et al. Frequency of influenza H3N2 intra-subtype reassortment: attributes and implications of reassortant spread. *BMC Biol* 2016; **14**: 117.
- Powell H, Pekosz A. Neuraminidase antigenic drift of H3N2 clade 3c.2a viruses alters virus replication, enzymatic activity and inhibitory antibody binding. *PLoS Pathog* 2020; **16**: e1008411.
- Xu X, Blanton L, Elal AIA, et al. Update: influenza activity in the United States during the 2018–19 season and composition of the 2019–20 influenza vaccine. *MMWR Morb Mortal Wkly Rep* 2019; **68**: 544–51.
- Flannery B, Kondor RJG, Chung JR, et al. Spread of antigenically drifted influenza A(H3N2) viruses and vaccine effectiveness in the United States during the 2018–2019 season. *J Infect Dis* 2020; **221**: 8–15.
- Ferdinands JM, Gaglani M, Ghamande S, et al. Vaccine effectiveness against influenza-associated hospitalizations among adults, 2018–2019, US Hospitalized Adult Influenza Vaccine Effectiveness Network. *J Infect Dis* 2021; **224**: 151–63.